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MILITARY SPECIFICATION

DISTRIBUTOR, WATER, COLLAPSIBLE-TANK: 900-GALLON CAPACITY, WITH SPRAYBAR ASSEMBLY

This specification is approved for use by the USA Belvoir Research and Development Center, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a 900-gallon collapsible water tank with spraybar and all necessary accessories and connections required for installation in the bed of a military, 2-1/2-ton, air-droppable dump truck, and 5 ton dump truck.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified, the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

L-P-378	- Plastic Sheet and Strip, Thin Gage, Polyolefin.
PPP-B-601	- Boxes, Wood, Cleated-Plywood.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: USA Belvoir Research and Development Center, ATTN: STRBE-DS, Fort Belvoir, VA 22060-5606 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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PPP-B-636 - Boxes, Shipping, Fiberboard.
PPP-C-843 - Cushioning Material, Cellulosic.
PPP-T-60 - Tape: Packaging, Waterproof.

MILITARY

MIL-P-116 - Preservation, Methods of.
MIL-T-704 - Treatment and Painting of Materiel.
MIL-S-7916 - Sealing Compound, Thread and Gasket, Fuel, Oil, and Water Resistant.
MIL-G-20241 - Gasket Material, Wool Felt, Impregnated, Adhesive, Pressure-Sensitive.
MIL-S-81733 - Sealing and Coating Compound, Corrosion Inhibitive.

STANDARDS

FEDERAL

FED-STD-H28 - Screw-Thread Standards for Federal Services.
FED-STD-191 - Federal Standard for Textile Test Methods.

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-130 - Identification Marking of U.S. Military Property.
MIL-STD-177 - Rubber Products, Terms for Visible Defects of.
MIL-STD-889 - Dissimilar Metals.
MIL-STD-1186 - Cushioning, Anchoring, Bracing, Blocking and Waterproofing; with Appropriate Test Methods.

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein.

DRAWINGS

ME

D13202E2870 - Repair Sealing Clamp Assembly 3-, 5-, and 7-1/2-Inch Sizes.

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FOOD AND DRUG ADMINISTRATION (FDA)

Code of Federal Regulations, Title 21, Chapter 1, Part 177 - Indirect Food Additives: Polymers, and Part 182 - Substances Generally Recognized as Safe.

(The Code of Federal Regulations (CFR) and the Federal Register (FR) are for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402. When indicated, reprints of certain regulations may be obtained from the Federal agency responsible for issuance thereof.)

(Copies of specifications, standards, and drawings required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

UNIFORM CLASSIFICATION COMMITTEE, AGENT

Uniform Freight Classification.

(Application for copies should be addressed to the Uniform Classification Committee, ATTN: Tariff Publishing Officer, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

NATIONAL MOTOR FREIGHT TRAFFIC ASSOCIATION, INC., AGENT

National Motor Freight Classification.

(Application for copies should be addressed to the American Trucking Associations, Inc., ATTN: Traffic Department, 1616 P. Street N.W., Washington, DC 20036.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS

ASTM D 412 - Rubber Properties In Tension.
ASTM D 573 - Rubber-Deterioration In An Air Oven.
ASTM D 750 - Rubber-Deterioration In Carbon Arc or Weathering Apparatus.
ASTM D 751 - Coated Fabrics.
ASTM D 2137 - Rubber Property - Brittleness Point of Flexible Polymers and Coated Fabrics.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

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2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Description. The water distributor shall consist of a collapsible, rubber-coated fabric tank, a gravity-fed spraybar, and accessories required to mount the distributor on a 2-1/2-ton, 6 x 6, M342A2, dump truck (see figure 1), or 5 ton, 6 x 6, M929 and M930 dump truck (see figure 2) as specified (see 6.2). With the tank containing 750 gallons of water the assembled distributor shall be capable of being transported at speeds up to 35 miles per hour (mph) over smooth-surfaced roads and at speeds up to 15 mph over unimproved roads without damage (see 4.6.2.7). The assembled distributor shall discharge water through all holes in the spraybar when the vehicle is moving forward at a speed of not more than 5 mph.

3.2 First article. When specified (see 6.2), a sample shall be subjected to first article inspection (see 4.4 and 6.3).

3.3 Material. Material shall be as specified herein. Material not specified shall be selected by the contractor and shall be governed by all the provisions of this specification. The finished tank shall contain no materials nor substances which might leak out or disintegrate and cause the water to become toxic. The material shall have no adverse affect on the health of personnel when used for its intended purpose. The liners and all surfaces that contact water shall be non-toxic and shall conform to Food and Drug Administration, Code of Regulations, Title 21, Chapter 1, Parts 177 and 182. Questions pertinent to this specification shall be referred by the procuring activity to the departmental medical service who will act as advisor to the procuring agency.

3.3.1 Material deterioration and control. The distributor shall be fabricated from compatible materials, inherently corrosion and deterioration resistant or treated to provide protection against the various forms of corrosion and deterioration that may be encountered in any of the applicable storage and operating environments to which the distributor may be exposed (see 3.7).

3.3.1.1 Dissimilar metals. Dissimilar metals, as defined in MIL-STD-889, shall be electrically insulated from one another to minimize or prevent galvanic corrosion. Insulation may be provided by an insulating barrier such as a corrosion inhibiting sealant conforming to MIL-S-81733 or chromate tape conforming to MIL-G-20241. Protection against corrosion could also be obtained by exclusion of electrolyte if feasible.

3.3.1.2 Identification of materials and finishes. The contractor shall identify the specific material, material finish or treatement for use with components and subcomponents, and shall make information available upon request, to the contracting officer or designated representative.

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3.3.2 Tank material.

3.3.2.1 Nylon cloth. Nylon cloth shall be of high tenacity and shall be heat set and scoured. After the cloth is heat set and scoured, physical characteristics shall be as follows:

Weight	- 7.25 to 8.50 oz./sq. yd.
Thickness	- 0.016 to 0.020 in.
Thread count	- 33 per in. for warp and 31 per in. for fill (min).
Residual shrinkage	- 1 percent (max).

Defects in cloth such as bagginess or uneven warp tension, tight or drawn selvage, smashes, oil spots, gouts, kinky fill, floats, shuttle marks, two continuous threads missing, soiling, or tears in selvage shall be corrected, cut out, or marked for future elimination.

3.3.2.2 Rubber compound. All rubber for coating the nylon cloth and for gum strips and sheets shall be formulated from a copolymer of butadiene and styrene. The use of high grade natural rubber to the limit of 20 percent of the copolymer is acceptable; the grade of rubber used shall comply with the requirements of 3.4. Water-absorption and water-extractable ingredients shall be held to a minimum. The physical characteristics of the vulcanized rubber shall be as follows:

Tensile strength, psi (min)	1,500
Ultimate elongation, percent (min)	300

3.3.2.3 Physical characteristics after accelerated weathering and aging. When subjected to accelerated weathering and aging, the vulcanized rubber compound shall retain not less than 75 percent of the original tensile strength and not less than 65 percent of the original ultimate elongation.

3.3.2.4 Physical characteristics at -40° F. The vulcanized rubber compound shall not crack at -40° F ±2° F.

3.3.2.5 Gum strips and sheets. Gum strips and sheets shall be between 0.015 inch and 0.030 inch thick before application.

3.3.2.6 Cement. Cement used in fabrication shall be compatible with the rubber compounds and shall be of a vulcanizing type cured by heat and pressure.

3.3.2.7 Coated cloth. The coated cloth shall consist of nylon cloth, calendar-coated on both sides with rubber compound. An adhesive may be used to obtain the required coating adhesion. One side shall be coated with a 0.012-inch- to 0.014-inch-thick coat, and the other side shall be coated with two coats having a total thickness of 0.018 inch to 0.020 inch. The coverage afforded by each coat shall be complete and nonporous. The double-coated side shall be identified by markings at regular intervals. After calendering, the coated cloth shall be rolled in liners. Not more than 40 days shall elapse between mixing and vulcanizing. When vulcanized, the coated cloth shall have the following physical properties:

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a. The breaking strength in both the fill and warp directions shall be not less than 350 pounds per inch of width. After the coated cloth has had 300 hours of accelerated weathering, the breaking strength of the coated cloth shall be not less than 90 percent of the original breaking strength.
b. The coated cloth shall withstand not less than 4,000 creasings without cracking. After 300 hours of accelerated weathering, the coated cloth shall withstand not less than 3,500 creasings without cracking.
c. The tearing strength shall be not less than 30 pounds in both the warp and fill directions.
d. The adhesion between the nylon cloth and the rubber compound shall be not less than 15 pounds per inch of width for the coated cloth before and after immersion in water for 14 days at a temperature of 158° F.
e. When overlapped as specified in 4.6.2.3.8, the coated cloth shall separate within 5 seconds under a 4-ounce load.

3.3.3 Recovered materials. For the purpose of this requirement, recovered materials are those materials which have been collected from solid waste and reprocessed to become a source of raw materials, as distinguished from virgin materials. The components, pieces and parts incorporated in the distributor may be newly fabricated from recovered materials to the maximum extent practicable, provided the distributor produced meets all other requirements of this specification. Used, rebuilt or remanufactured components, pieces and parts shall not be incorporated in the distributor.

3.4 Design and construction.

3.4.1 Collapsible water tank. The collapsible water tank shall have a capacity of not less than 900 gallons. The fabrication tank and fittings shall be watertight.

3.4.1.1 Tank construction. The tank body shall be fabricated from uncured cloth with the double-coated side of the cloth placed on the inside of the tank. All cut edges and stitching shall be sealed with strips or sheets or with rubber cement. The tank shall be fabricated so as to have 4-ply material thickness on the bottom surface and 6 inches, ±0.5 inch, up from the bottom of the tank around the perimeter. Step-down from 4-ply to single-ply thickness shall be accomplished around the perimeter of the tank within a 10-inch height measured from the base of the tank. All seams shall be not less than 3 inches wide and shall develop the full breaking strength of the coated cloth. Cemented areas shall not slip or fail when the tank is folded or while supporting a load of 40 pounds per inch of width at a temperature of 160° F for 22 hours. The tank shall be fabricated so as to be capable of being mounted in the bed of the 2-1/2-ton dump truck and 5 ton dump truck (see 3.1). The tank shall have the configuration and dimensions as shown in figure 3. The tank shall be provided with an aluminum manhole flange attachment conforming to figure 7, located at the top of the tank as shown in figure 3. A water-tight aluminum cover shall be provided and shall be bolted to the flange. Provision shall be made to fill the tank through a 2-inch NPSH threaded male aluminum fitting, 11-1/2 threads per

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inch, in accordance with FED-STD-H28, bolted to the flange cover. An aluminum, full-opening strainer, with perforations smaller than the spraybar holes, shall be provided to filter the water while the tank is being filled. A threaded aluminum vent-pipe not less than 1/2 inch in diameter by 2-1/2 inches long shall be incorporated into the flange cover. Caps shall be provided for both the fill and vent pipes. Retaining chains shall be provided to prevent loss of the caps. All fittings shall be watertight when assembled.

3.4.1.2 Tank fittings. Fittings shall be installed before the tank is cured to prevent leakage between the fitting and the tank wall. Molded fittings shall not be adversely affected by the additional cure necessary when they are vulcanized with the tank body.

3.4.1.3 Repairs. Defects incurred prior to curing shall be corrected prior to curing. Such corrections shall not be considered as repairs. Damages occurring after curing, which involve reducing the strength of the supporting fabric, shall be repaired by the hot-patch method with coated fabric as used in the tank. Such repairs shall extend a minimum of 3 inches beyond any damaged areas and the cord ends shall be sealed with strips as specified in 3.4.1.1. Repairs shall restore the full strength of the coated cloth. Repair patches shall have an adhesion of not less than that specified in 3.3.2.7. Where possible, a puncture repair shall consist of patches on both sides of the coated cloth. Not more than six repairs for each tank body shall be permitted.

3.4.1.3.1 Exposed nylon cloth. Abraded, damaged, or bald spots which exposes but does not injure the nylon cloth of the coated fabric of the tank shall be corrected by hot-plating a gum strip or gum sheet over the exposed area and shall be counted as a repair.

3.4.1.3.2 Seam leaks. Seam leaks which form droplets or run down or form wet spots greater than 4 inches in diameter in 5 hours shall be repaired by the hot-patch method specified in 3.4.1.3 and shall count as repairs.

3.4.1.3.3 Blisters. Blisters or nonadhered areas between plies of laminated fabric involving an area smaller than 4 inches in diameter or a similiar area with a maximum dimension not greater than 12 inches in any direction shall not require opening, provided that a 1-1/4-inch minimum width continuous effective bond exists around the blister. Blisters between plies of laminated fabric which do not fall within the above limits shall be opened, buffed, cemented and hot-plated. If it is necessary to cut the fabric or if the blister is located in a tank seam, such cut or tank seam blisters shall be patched by the hot-patch method specified in 3.4.1.3 and counted as a repair. Blisters between the coating compound and the base fabric with the largest dimension over 2 inches shall be corrected by removing loose gum and hot-plating in accordance with 3.4.1.3.

3.4.1.3.4 Nonadhered areas. Nonadhered areas exceeding the limits specified in 3.13a shall be buffed, cemented, and hot-plated. This shall not be counted as a repair unless involving a tank seam that does not provide a 2-1/2-inch continuous bond overlap, in which event a hot-patch shall also be applied and counted as a repair.

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3.4.1.4 Curing. After fabrication, the tank shall be vulcanized in a steam or air pressure autoclave at a gage pressure of not less than 40 psi and at a temperature of not less than 260° F. An adhesive sample shall be provided with each cure.

3.4.2 Spraybar. The spraybar shall be located at the rear of the tank as shown in figure 1 or figure 2. The spraybar shall consist of a center section incorporating provisions to add an extension to each end of the spraybar to make a total spraybar length of 16 feet. The spraybar shall be capable of distributing water through the center section and both extensions. The spraybar shall fold to a length not greater than that of the center section. A retaining chain shall be provided to hold the spraybar in the folded position. The spraybar shall be assembled as shown in figure 4 or figure 5. Water shall flow into the spraybar through a 2-inch fitting attached to the rear of the tank. Where the fitting projects into the tank, protection of the tank bottom shall be provided to prevent damage to the tank when the fitting contacts the tank bottom during folding of the tank. There shall be no evidence of damage or leakage when the discharge fitting is tested as specified in 4.6.2.6.3. The flow of water shall be controlled by a 2-inch brass or bronze self-closing lever valve which shall be reliably operated by means of a nylon rope from the cab of the dump truck. The end of the rope shall be equipped with a wooden handle that is 4 ±1/4 inches long and 1/2 to 3/4 inch in diameter. The center of the handle shall be drilled and the nylon rope shall be secured to the handle by knots on both sides of the handle. The handle shall be tapered at both ends. Union(s) shall be used to insure correct alignment of the self closing valve. The fittings shall be assembled using joint sealant so that the assembled spraybar shall be watertight at all points of connection. The joint sealant shall conform to MIL-S-7916. The spraybar shall be supported by means of two steel braces. The braces shall be fabricated as shown in figure 6. Two 1/2-inch diameter steel U-bolts shall be furnished for bolting the braces to the tailgate lugs of a dump truck. Washers and lock nuts shall also be provided.

3.5 Emergency repair items. The emergency repair items listed in table I shall be furnished with each distributor. The technical manual shall identify all repair items and procedures for emergency repair.

TABLE I. Emergency repair items.

Item	Quantity	Description
Clamp, sealing, 3-in.	2	D13202E2870
Clamp, sealing, 5-in.	1	D13202E2870
Clamp, sealing, 7-in.	3	D13202E2870

3.5.1 Drawing. The drawing forming a part of this specification is an end product drawing. No deviation from the prescribed dimensions or tolerances is permissible without prior approval of the contracting officer. Where tolerances could cumulatively result in incorrect fits, the contractor shall provide tolerances within those prescribed on the drawings to insure correct fit, assembly and operation of the repair sealing clamp assembly 3-, 5-, and 7-1/2-inch sizes. Any data (e.g., shop drawings, layouts, flow sheets, processing procedures, etc.)

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prepared by the contractor or obtained from a vendor to support fabrication and manufacture of the production item shall be made available, upon request for inspection by the contracting officer or the designated representative.

3.6 Leakage. The finished tank shall not leak in the seam areas. Weepage through the tank body which does not form droplets and run down the sides of the tank and does not produce a wet spot greater than 6 inches in diameter during the 5-hour testing period (see 4.6.2.6.2) will be acceptable. The total area of wet spotting shall not exceed 1-1/2 square feet per tank. During the 1 hour test (see 4.6.2.6.2) the above requirements shall apply, except that the maximum size wet spot shall be not more than 3 inches in diameter, and total area of wet spots shall not exceed 45 square inches. Leaks in the tank body, weepage that forms droplets that run down the sides of the tank, or wet spots that exceed the above limits shall be repaired by the hot-patch method specified in 3.4.1.3 and shall count as a repair.

3.7 Environmental requirements. The distributor shall conform to the following environmental requirements.

3.7.1 Operating temperature. The distributor shall perform as specified in 3.1 in any ambient temperature from +120° F to +33° F.

3.7.2 Storage temperature. The distributor shall not be damaged by storage in any ambient temperature from +160° F to -40° F.

3.8 Repair parts and maintenance tools. Such repair parts and maintenance tools as are specified (see 6.2) shall be furnished.

3.9 Items to be furnished with the distributor. No shipment of distributors shall be made unless repair parts, tools, and accessories are included with the shipment or unless approval for shipment of the distributors without such items has been received from the contracting officer.

3.9.1 Accessories. Twenty-five feet of 2-inch, unlined linen hose shall be provided with each tank. The hose shall be fitted on both ends with female swivel fittings. All threads shall be NPSH. The hose shall be used for filling the tank. A 90-degree street elbow shall be provided to prevent kinking of the hose at the tank inlet while the tank is being filled. Threads shall be NPSH.

3.10 Treatment and painting. All metal components of the distributor shall be cleaned, treated, and painted in accordance with MIL-T-704, type B. The color of the distributor shall be lusterless black.

3.11 Identification marking. The distributor shall be identified in accordance with MIL-STD-130.

3.11.1 Stenciling. The gross weight of the distributor and the capacity of the tank shall be stenciled on each side of the tank in letters not less than 1 inch high. Stenciling shall be in accordance with MIL-T-704.

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3.12 Government-loaned property. Unless otherwise specified, the following property as applicable in the quantities indicated will be loaned by the Government (see 6.4):

Item	Description	Identification	Quantity
1	Truck, dump, 5-ton, 6 x 6, M929	NSN 2320-01-047-8756	1
	or		
2	Truck, dump, 5-ton, 6 x 6, M930	NSN 2320-01-047-8755	1
	or		
3	Truck, dump, 2-1/2-ton, 6 x 6, M342A2	NSN 2320-00-077-1643	1

3.13 Workmanship. The distributor shall be cleaned free of dirt, extraneous talc, back-up paper or cloth, excess material used in fabrication of tanks, and other foreign materiel. Stains occurring as a result of the vulcanization shall not be considered as dirt or foreign material. All parts shall be clean, free from rust, toolmarks, pits and other injurious defects. External surfaces shall be free of burrs, sharp edges and corners except where sharp edges or corners are required and they are not detrimental to safety. The cured tank body shall be free of the following defects as defined in MIL-STD-177:

a. Poor or insufficient adhesion (defined as those areas that exceed 1/8-inch nonadhesion at the edges for tank water holding seams or 1/2 inch for other cemented components).
b. Any condition that exposes the nylon cloth of the coated fabric; bald spots, abrasions.
c. Any condition that damages the nylon cloth not repaired (cuts, tears, punctures).
d. Buckles, puckers, or wrinkles in seams over 1/4-inch width that extend over 1/2 inch from seam edge, or that reduces flat contact to less than 2-1/2 inches for splices and seams 3 inches or over in width.
e. Blisters (between the elastomeric coating and the nylon fabric with any dimension exceeding 2 inches). A blister between plies and laminated construction involving an area less than 4 inches in diameter is acceptable provided the maximum linear dimension does not exceed 12 inches. A total number of blisters in lamination exceeding 5 per square foot is unacceptable.
f. Any other defect that will adversely affect the strength or serviceability of the items.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified

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herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all section 3 and 5 requirements. The inspections in section 4 are the minimum to be used to demonstrate compliance. Sampling in quality conformance inspection does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4).
b. Quality conformance inspection (see 4.5).
c. Inspection of packaging (see 4.7).

4.3 Toxicological data and formulations inspection. The contractor shall furnish to the contracting officer a certificate of compliance that the liners and all surfaces that contact the water complies with the Code of Federal Regulations, Food and Drugs. The certificate shall be supplied by an independent material testing laboratory (see 3.3).

4.4 First article inspection.

4.4.1 Examination. The distributor shall be examined as specified in 4.6.1. Presence of one or more defects shall be cause for rejection.

4.4.2 Tests. The distributor or representative material of the distributor, as applicable shall be tested as specified in 4.6.2.1 through 4.6.2.7. Failure of any test shall be cause for rejection.

4.5 Quality conformance inspection.

4.5.1 Sampling.

4.5.1.1 Nylon cloth. A sample of the nylon cloth not less than 1-1/2 feet long by the full width of the roll shall be selected at random from each 2,000 linear yards or fraction thereof.

4.5.1.2 Rubber compound. A sample of the rubber compound of sufficient size to perform the required tests shall be furnished from each batch or lot of 4,000 pounds or fraction thereof.

4.5.1.3 Coated cloth. A sample of the coated cloth not less than 1-1/2 feet long by the full width of the roll shall be selected from each 2,000 linear yards or fraction thereof of uncured coated cloth.

4.5.1.4 Seams. One sample seam shall be furnished for each five tanks produced.

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4.5.2 Examination. Each distributor shall be examined as specified in 4.6.1. Presence of one or more defects shall be cause for rejection of the applicable distributor.

4.5.3 Tests.

4.5.3.1 Individual. Each tank shall be tested as specified in 4.6.2.5 and 4.6.2.6. Failure of any test shall be cause for rejection.

4.5.3.2 Samples. Samples selected in accordance with 4.5.1 shall be subjected to the tests specified in 4.6.2.1 through 4.6.2.3.2, 4.6.2.3.4, and 4.6.2.3.6 through 4.6.2.4.2, as applicable. Failure of samples of nylon cloth, rubber compound, coated cloth, or seams to pass any test shall be cause for rejection of material represented by the sample and rejection of any distributors that were fabricated from the rejected materials.

4.6 Inspection procedure.

4.6.1 Examination. The distributor shall be examined for the following defects:

101. Dimensions not as specified.
102. Assembly not complete or not correct.
103. Material not as specified.
104. Materials are not resistant to corrosion or deterioration or treated to be made resistant to corrosion or deterioration for the applicable storage and operating environment.
105. Dissimilar metals as defined in MIL-STD-889 are not effectively insulated from each other.
106. Contractor does not have documentation available for identification of material, material finishes or treatments.
107. Tank construction or tank fittings not as specified.
108. Spraybar not as specified or handle not as specified.
109. Mounting accessories not as specified.
110. Emergency repair items not as specified or missing.
111. Identification marking missing or not as specified.
112. Treatment, painting, or stenciling not as specified.
113. Repair parts or maintenace tools missing or not as specified.
114. More than six repairs per tank.
115. Workmanship not as specified.
116. Certificate of compliance for FDA approved liners and surfaces not available.

4.6.2 Tests.

4.6.2.1 Nylon cloth. The thickness, weight, and thread count of the nylon cloth shall be determined in accordance with FED-STD-191, methods 5030, 5040, and 5050, respectively. Residual shrinkage after heat set and scouring shall be measured by exposing the specimen in an open oven for 45 minutes at a temperature of 300° F under a tension of 2 ounces per inch of width of specimen. Nonconformance to 3.3.2.1 shall consitute failure of this test.

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4.6.2.2 Rubber compound. The sample shall be vulcanized identically with the tanks fabricated from the rubber compound represented by the sample. After vulcanization, the sample shall be 1 foot square and 0.065 inch to 0.080 inch thick.

4.6.2.2.1 Tensile strength and ultimate elongation. The tensile strength and ultimate elongation of rubber compound shall be determined in accordance with ASTM D 412, using die C specimens. Nonconformance to 3.3.2.2 shall constitute failure of this test.

4.6.2.2.2 Tensile strength and ultimate elongation after accelerated weathering. Accelerated weathering shall be in accordance with ASTM D 750, using the light-aging method with the following exceptions. Buff all specimens on both sides and stretch to 10 percent elongation within 3 hours after buffing. Condition the stretched specimens for 16 to 24 hours at 100° F ±5° F immediately after racking to facilitate blooming of protective waxes. Start the exposure in the light-aging unit within 1 hour after removal from the 100° F conditioning oven. Exposure for 100 hours shall be made at 10 percent elongation using "Eveready Sunshine" carbons or equal. Remove four of the eight "Corex D" filters leaving every alternate glass in place. Total dosage in decomposition of oxalic acid by uranyl oxalate actinometer shall be not less than 20,000 mg per square decimeter. Determine the tensile strength and ultimate elongation of the rubber compound after accelerated weathering in accordance with ASTM D 412 using die C specimens. Nonconformance to 3.3.2.3 shall constitute failure of this test.

4.6.2.2.3 Tensile strength and ultimate elongation after accelerated aging. Accelerated aging shall be in accordance with ASTM D 573. Determine the tensile strength and ultimate elongation of rubber compound after accelerated aging in accordance with ASTM D 412, using die C specimens. Specimens shall be aged for 166 hours ±1 hour at 70° C (158° F). Nonconformance to 3.3.2.3 shall constitute failure of this test.

4.6.2.2.4 Brittleness at -40° F. Test specimens shall be T-50 having a section 0.080 inch wide by 0.080 inch thick (maximum) by 1-1/2 inches long plus a 1/4-inch square stub of the same thickness at each end. Three specimens shall be used for each set of tests. Condition the specimens for not less than 168 hours in a chamber maintained at -40° F ±3.6° F. Condition the test apparatus for not less than 2 hours at -40° F ±3.6° F. Test the specimens in accordance with ASTM D 2137, method B. When the specimen must be transferred for test, their temperature shall not be raised above -25° F. Use tongs that have been conditioned prior to use for not less than 1 hour at -40° F ±2° F for handling test specimens. Cracking or breaking of the test specimens shall constitute failure of this test. Nonconformance to 3.3.2.4 shall constitute failure of this test.

4.6.2.3 Coated cloth. The sample shall be vulcanized identically with the tanks fabricated from the coated cloth represented by the sample.

4.6.2.3.1 Coating thickness. Determine the coating thickness with a micrometer based on the following criteria:

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a. When the total thickness of nylon cloth coated with one coat of rubber compound is less than 0.028 inch or more than 0.034 inch, consider the thickness of the coating as less than 0.012 inch and greater than 0.014 inch, respectively.
b. When the total thickness of the coated cloth is less than 0.046 inch or greater than 0.054 inch, consider the thickness of the coating as less than 0.018 inch and greater than 0.020 inch respectively.

Nonconformance to 3.3.2.7 shall constitute failure of this test.

4.6.2.3.2 Breaking strength. Determine the breaking strength of coated cloth in accordance with FED-STD-191, method 5102. Insert pieces of cotton duck or blotting paper between the specimens and the faces of each jaw to prevent the jaws from cutting the test specimen. Nonconformance to 3.3.2.7 shall constitute failure of this test.

4.6.2.3.3 Breaking strength after accelerated weathering. Subject the coated cloth specimens to 300 hours of accerated weathering in accordance with FED-STD-191, method 5804, except that alternate filters shall be removed. Determine the breaking strength in accordance with FED-STD-191, method 5102. Insert pieces of cotton duck or blotting paper between the specimens and the faces of each jaw to prevent the jaws from cutting the test specimen. Nonconformance to 3.3.2.7 shall constitute failure of this test.

4.6.2.3.4 Flexibility. Fold a 1- by 4-inch specimen from a flat position to a closed U-shaped position in movable jaws set at an opening of 2.4 times the thickness of the specimen. The movable jaws shall be parallel to each other and shall open and close 175 times a minute. Nonconformance to 3.3.2.7 shall constitute failure of this test.

4.6.2.3.5 Flexibility after accelerated weathering. Subject the coated cloth specimens to the accelerated weathering specified in 4.6.2.3.3, then test as specified in 4.6.2.3.4. Nonconformance to 3.3.2.7 shall constitute failure of this test.

4.6.2.3.6 Adhesion. Bond two layers (plies) of uncured coated cloth (not less than 6 inches wide) together using the cement and technique used in forming seams in the construction of the tank body. Cure the assembled plies along with and identically to the tanks. Cut the cured, bonded, plied cloth into strips 2 inches wide by 6 inches long. Separate the ends of the plies for a length of about 1-1/2 inches by cutting through to the cloth. Determine the adhesion of rubber to nylon cloth in accordance with FED-STD-191, method 5970, except that rate of travel shall be 2 inches per minute $\pm$0.25 inch per minute. Repeat this test again after immersing the specimen in water for 14 days at a temperature of 160° F. Nonconformance to 3.3.2.7 shall constitute failure of this test.

4.6.2.3.7 Tearing strength. Determine the tearing strength of coated cloth in accordance with FED-STD-191, method 5134. Nonconformance to 3.3.2.7 shall constitute failure of this test.

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4.6.2.3.8 Blocking. Place two coated fabric specimens (6 inches by 1 inch) in an oven on a smooth surface in such a manner that the ends are overlapped 1 inch. Place a 4-pound weight directly on the overlapped area. After conditioning at a temperature of 160° F ±2° F for 4 hours, remove the weight and take the joined specimens from the oven and condition them for 1 hour at 73.4° F ±3.6° F. Fix one end of the specimen in a clamping device allowing the free end to hang. Suspend a 4-ounce load from the free end of the specimen. Nonconformance to 3.3.2.7 shall constitute failure of this test.

4.6.2.4 Seams. Sample seams shall be vulcanized identically with the tanks fabricated from the coated cloth represented by the sample.

4.6.2.4.1 Breaking strength. Determine the breaking strength of the seam in accordance with ASTM D 751. Nonconformance to 3.4.1.1 shall constitute failure of this test.

4.6.2.4.2 Slippage. Determine the seam slippage by suspending the specimen for 22 hours in an air-circulating oven with a 40-pound dead weight hanging freely from the unattached end. The temperature of the oven shall be 160° F. The test specimen shall be a strip 1 inch wide, cut completely across and extending 3 inches beyond the seam. Nonconformance to 3.4.1.1 shall constitute failure of this test.

4.6.2.5 Capacity. Fill the tank to maximum capacity with clean water. Measure the amount of water required to fill the tank. Inability of the tank to hold 900 gallons (see 3.4.1) shall constitute failure of this test.

4.6.2.6 Water leakage.

4.6.2.6.1 Leakage after 1 hour. The tank shall be erected on a flat, level surface covered with white absorbent paper and filled with water colored red or green with certified food coloring. The filled tank shall remain undistrubed for not less than 60 minutes and shall then be examined for leaks. Nonconformance to 3.6 shall constitute failure of this test. If leaks are detected, the leaking tanks shall be repaired in accordance with 3.4.1.3, and these tanks shall be retested.

4.6.2.6.2 Leakage after 5 hours. The tank shall be folded five times as specified in 5.1.1.6 and placed in an open shipping container for 4 hours at a temperature of not less than 68° F. At the end of the 4 hour period, the tank shall be unfolded and refolded five times. The tank shall be erected and placed in position on a flat, level surface covered with white absorbent paper. The tank shall be filled with water colored with red or green certified food coloring and left to stand for 5 hours. Nonconformance to 3.6 or 3.13 shall constitute failure of this test.

4.6.2.6.3 Discharge fitting. Suspend the empty tank by the discharge fitting (see 3.4.2) for a period of not less than 5 minutes. Lower the tank, fill to capacity, and allow to stand for not less than 1 hour. At the end of this period inspect the discharge fitting and the area around the fitting for damage or leakage. Evidence of damage or leakage shall constitute failure of this test.

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4.6.2.7 Road. Mount the water distributor with spraybar in the folded position on the Government-loaned dump truck (see 6.4). Fill the tank with 750 gallons of clear clean water. Drive the truck for 50 miles. Fifteen miles shall be over unimproved roads at speeds up to 15 mph and the remaining portion of the road test shall be over smooth-surfaced roads at speeds up to 35 mph while maintaining an average speed of 25 mph. At the end of the 50-mile test, extend the spraybar and discharge water through the spraybar while the truck is moving forward at a speed of not more than 5 mph. Seepage, leakage, or other damage shall constitute failure of this test.

4.7 Inspection of packaging.

4.7.1 Quality conformance inspection of pack.

4.7.1.1 Unit of product. For the purpose of inspection, a completed pack prepared for shipment shall be considered a unit of product.

4.7.1.2 Sampling. Sampling for examination shall be in accordance with MIL-STD-105.

4.7.1.3 Examination. Samples selected in accordance with 4.7.1.2 shall be examined for the following defects. AQL shall be 2.5 percent defective.

117.	Materials, methods, or containers not as specified for level A. Each incorrect material, method, or container shall be considered one defect.
118.	Spraybar not removed and disassembled as specified.
119.	Spraybar assembly and braces not cushioned as specified for level A.
120.	Components placed in boxes not cushioned as specified for level A
121.	Cotton hose not coiled and secured as specified for level A.
122.	Caps not installed on valves.
123.	Tank not folded as specified.
124.	Boxes not provided with skids for level A.
125.	Blocking and bracing not in accordance with the referenced standard for level A.
126.	Strapping not placed adjacent to the skids for level A.
127.	Marking missing, illegible, incorrect, or incomplete.

5. PACKAGING

5.1 Preservation. Preservation shall be level A, or C as specified (see 6.2).

5.1.1 Level A.

5.1.1.1 Spraybar assembly and braces. The spraybar assembly and braces shall be removed from the tank body. Disassembly of the spraybar shall be confined to that necessary to accomplish the preservation and packing specified herein. The components shall be wrapped in cushioning material conforming to PPP-C-843, type II, class B, and the cushioning secured in place with tape conforming to PPP-T-60, type IV.

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5.1.1.2 Emergency repair items. The emergency repair items shall be placed together in a close-fitting box conforming to PPP-B-636, W6c. The contents shall be cushioned to prevent movement and damage to the patching material with cushioning specified in 5.1.1.1.

5.1.1.3 Technical publications. Technical publications shall be preserved in accordance with MIL-P-116, method IC-1 or IC-3.

5.1.1.4 Repair parts and maintenance tools. The preservation application criteria and applicable methods of preservation of MIL-P-116 shall be used to preserve the repair parts and maintenance tools. The preserved components shall then be placed in a close-fitting box conforming to PPP-B-636, W6c. The contents shall be cushioned to prevent movement and damage with cushioning material specified in 5.1.1.1.

5.1.1.5 Cotton hose. The cotton hose shall be coiled and the coil secured with not less than four equally spaced ties of cotton tape or twine.

5.1.1.6 Tank. The tank shall be laid flat with the fittings up and the caps installed. The other preserved components shall be placed on top of the tank around the fittings and the tank folded over the components in a manner to form a compact bundle. Each tank shall be dusted with talc or other anti-sticking compound to prevent adhesion between folds. The talc or anti-sticking compound shall not make the water toxic.

5.1.2 Level C. The spraybar assembly and braces shall be removed. The caps shall be placed on the fittings and the tank folded as specified in 5.1.1.6. Preservation of the other distributor components shall be adequate to afford protection against deterioration and damage from the contractor to the initial destination.

5.2 Packing. Packing shall be level A, or C, as specified (see 6.2).

5.2.1 Level A. Each complete distributor shall be packed in a close-fitting box conforming to PPP-B-601, overseas type, grade B, style optional. The box shall be provided with skids. The contents shall be blocked and braced to prevent movement in accordance with MIL-STD-1186. The box shall be closed and strapped in accordance with the appendix to the box specification. Strapping shall be placed adjacent to the skids. The box shall be loosely lined with polyethylene conforming to L-P-378, type I, class 1, grade C, finish 1, 7 mil minimum thickness.

5.2.2 Level C. Each complete distributor shall be packed to assure carrier acceptance and safe delivery to destination at lowest rating in a container complying with Uniform Freight Classification rules or National Motor Freight Classification rules. The box shall be loosely lined with a 6 mil polyethylene material to prevent abrasion of the tank.

5.3 Marking. Marking for shipment shall be in accordance with MIL-STD-129.

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6. NOTES

6.1 Intended use. The collapsible tank water distributor is intended for use in dust alleviation and soil stabilization of roads and airfields.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification.
b. Whether the distributor will be mounted on 2 2-1/2- or 5-ton truck (see 3.1).
c. When first article is required (see 3.2)
d. Repair parts and maintenance tools required (see 3.8).
e. Degree of preservation and packing required (see 5.1 and 5.2).

6.3 First article. When a first article inspection is required, the item should be a preproduction model. The first article should consist of one distributor. The contracting officer should include specific instructions in the acquisition documents regarding arrangements for examinations, test, approval (or disapproval) of the first article test results, and disposition of the first article documents.

6.4 Government-loaned property. The contracting officer should arrange to lend the property listed in 3.12.

6.5 For Army procurement. The Surgeon General of the Army shall act as advisor in making health hazard determination.

Custodian:
Army - ME

Preparing activity:
Army - ME

Review activity:
DLA-CS

Project 3825-A168

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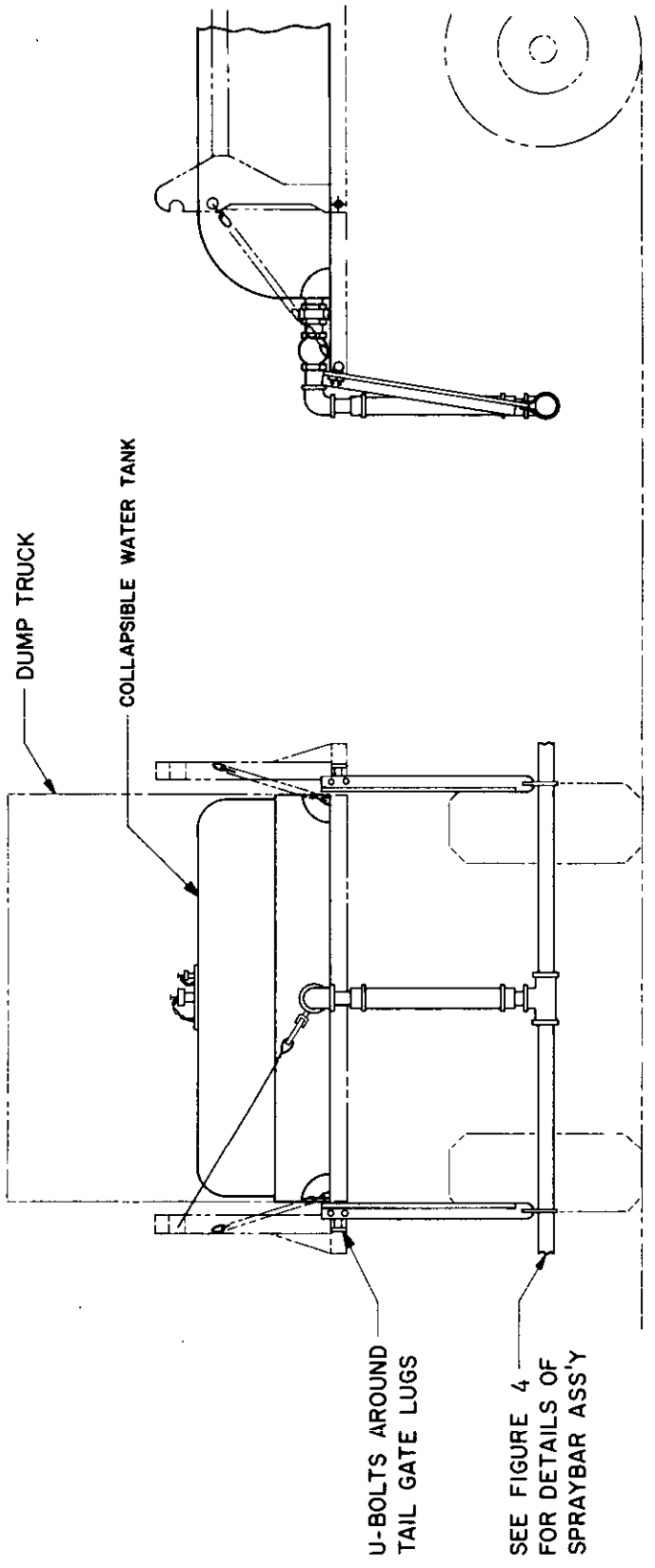
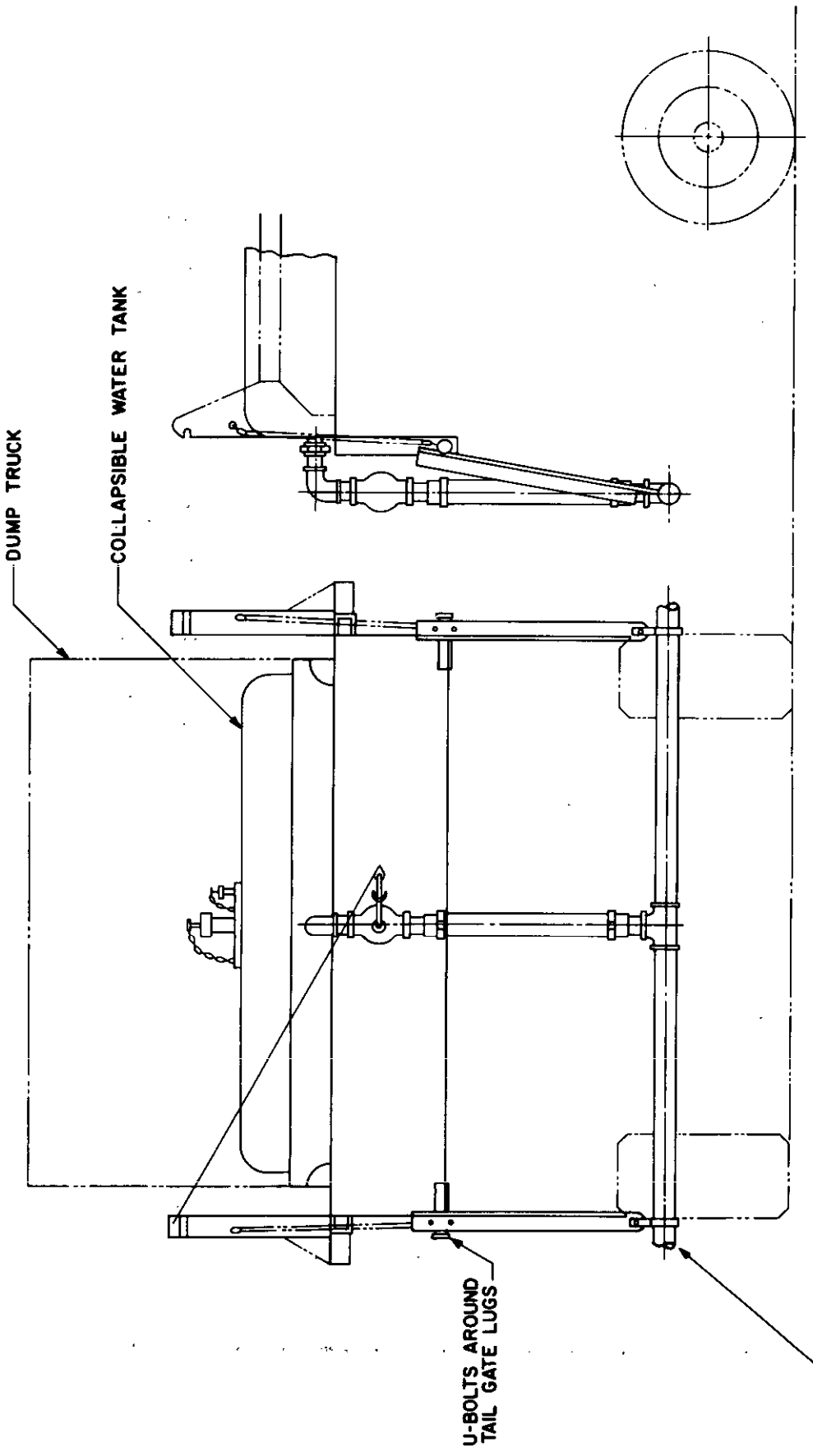


FIGURE 1. 900 GALLON COLLAPSIBLE WATER DISTRIBUTOR-MOUNTED ON BED OF 2 1/2 TON DUMP TRUCK

DX-1003A

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X-4290

FIGURE 2. 900 gallon collapsible water distributor - mounted on bed of 5 ton dump truck.

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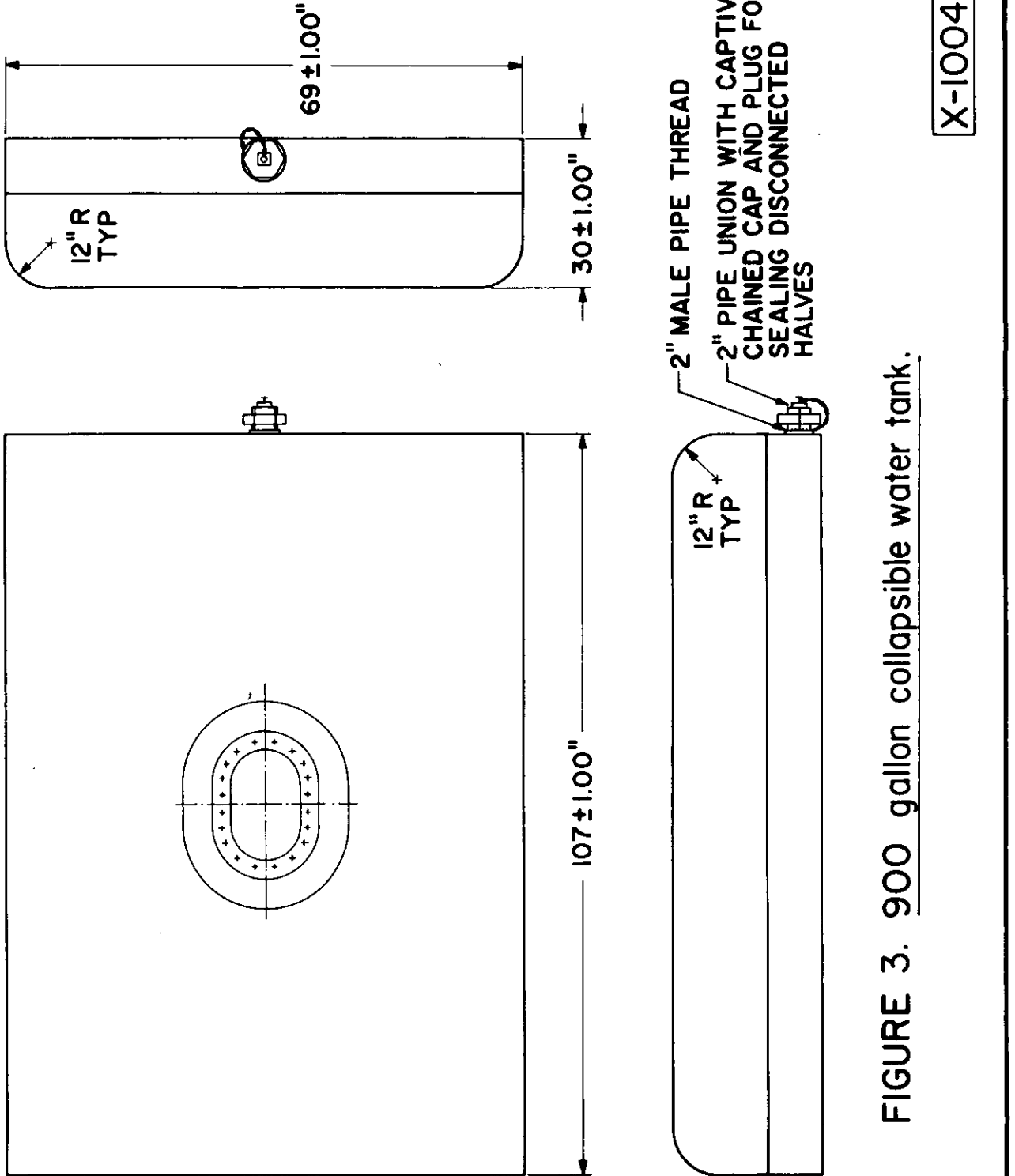


FIGURE 3. 900 gallon collapsible water tank.

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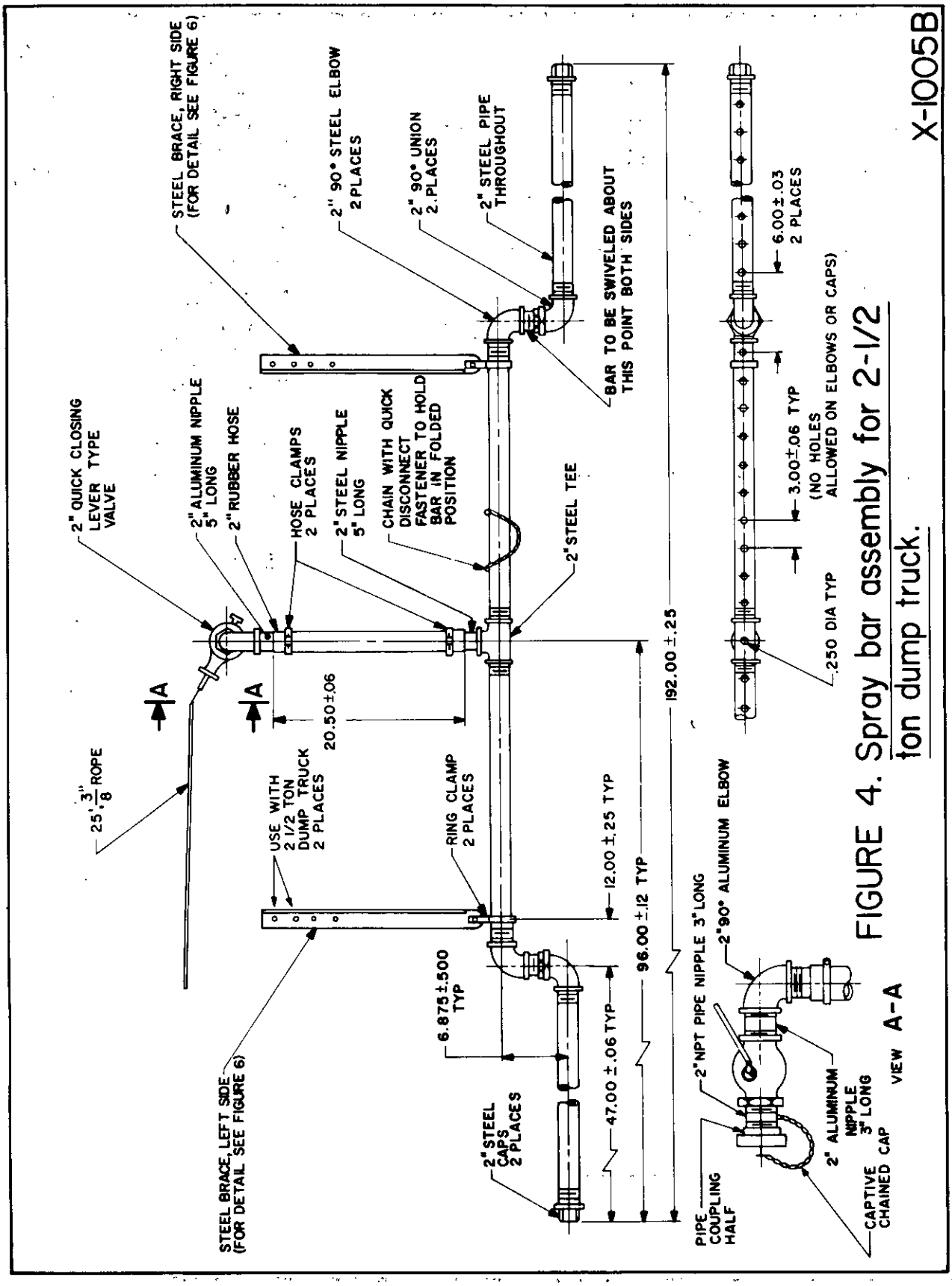
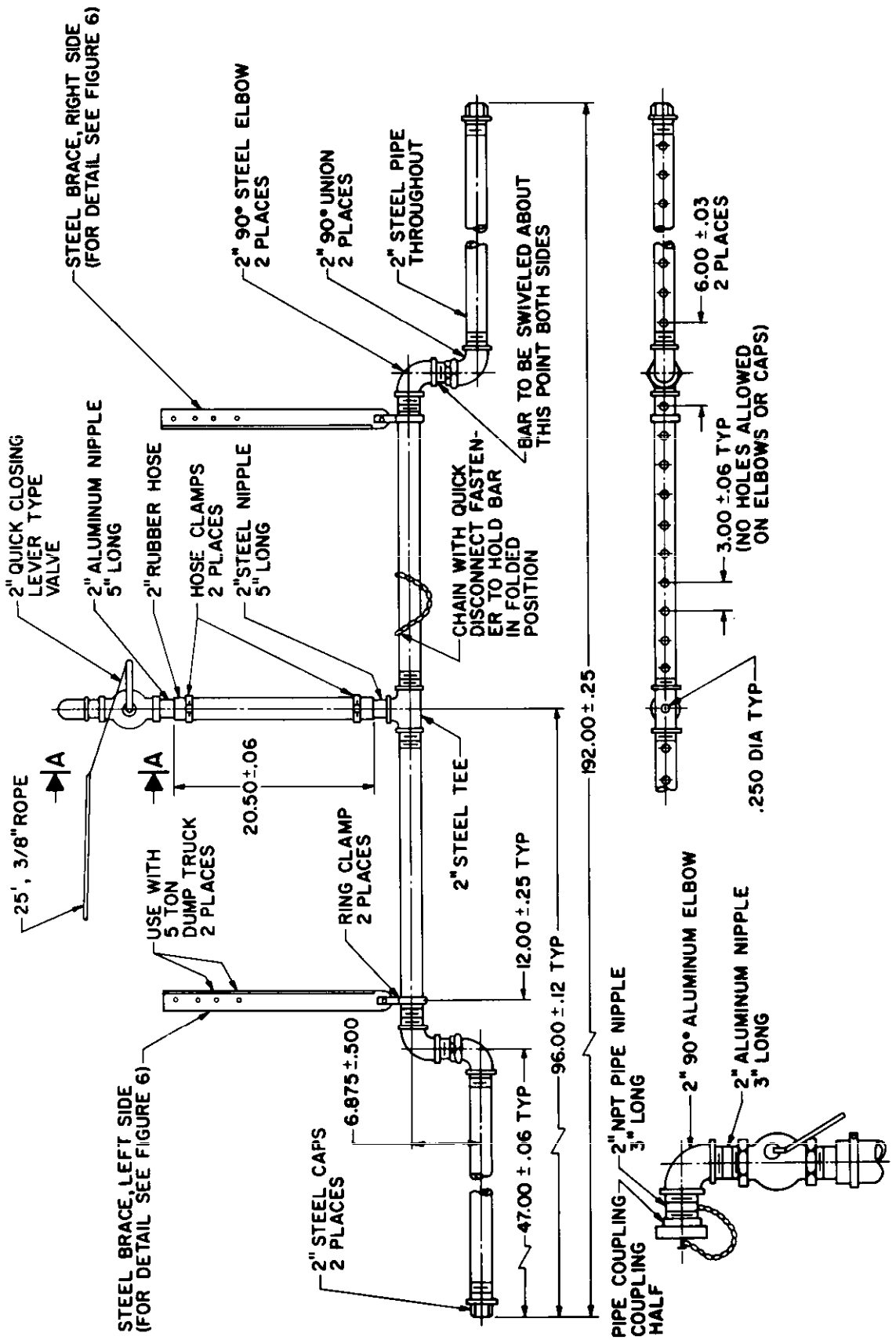


FIGURE 4. Spray bar assembly for 2-1/2 ton dump truck.

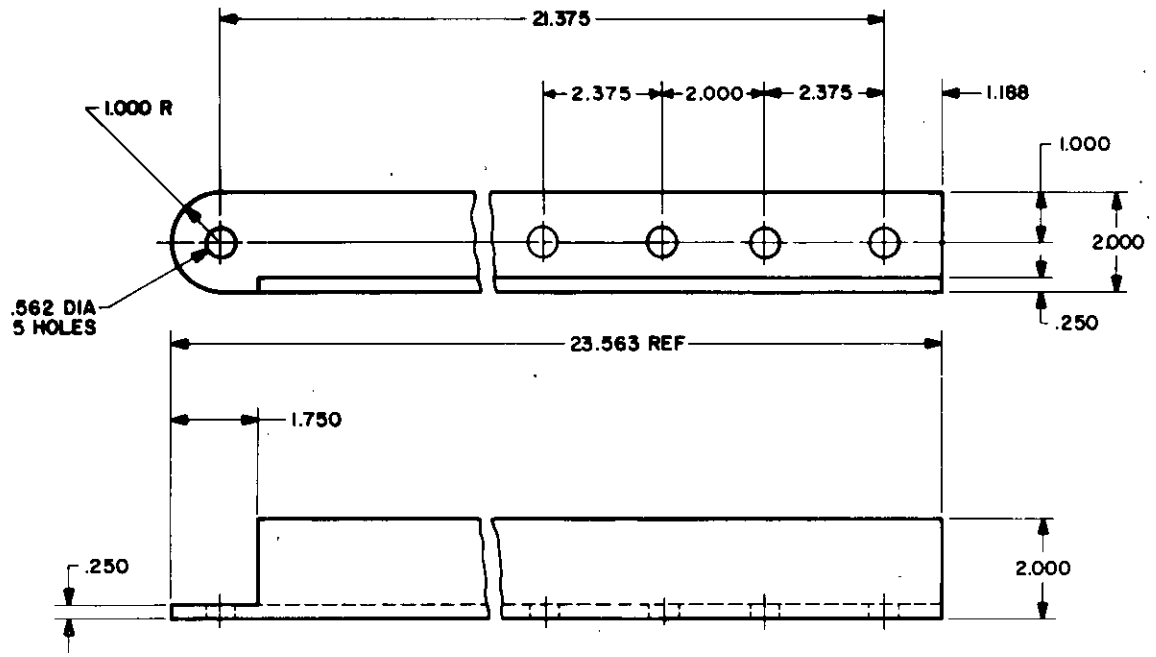
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X-4291

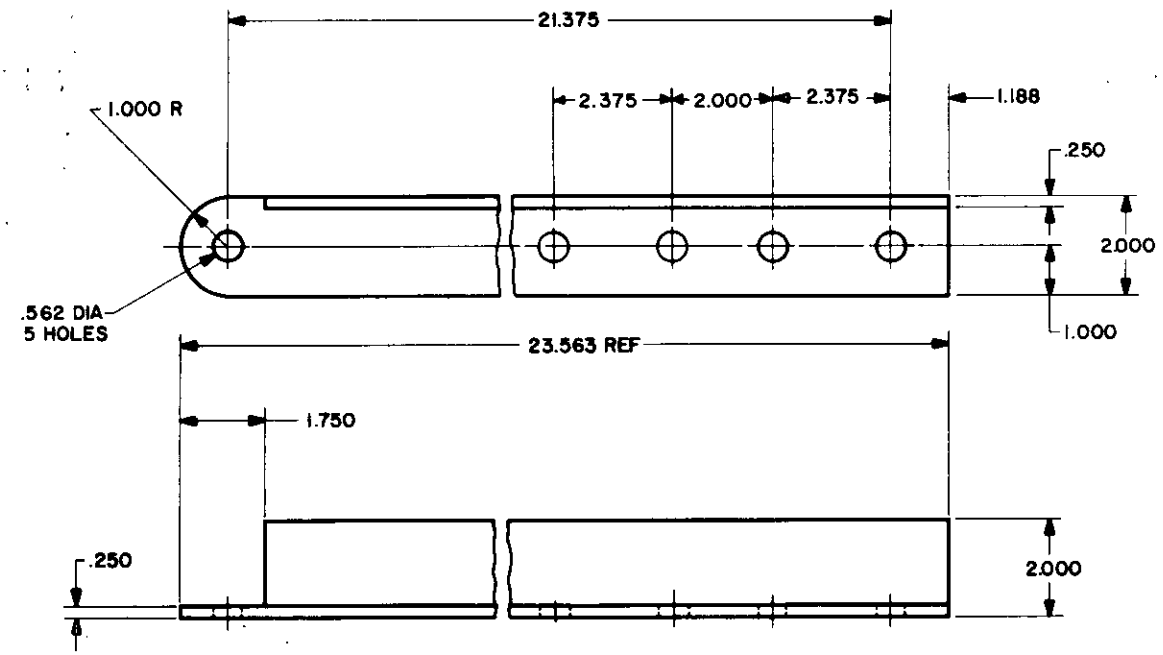
FIGURE 5. Spray bar assembly for 5 ton dump truck.

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ALL DIMENSIONS ARE IN INCHES

STEEL BRACE, LEFT SIDE



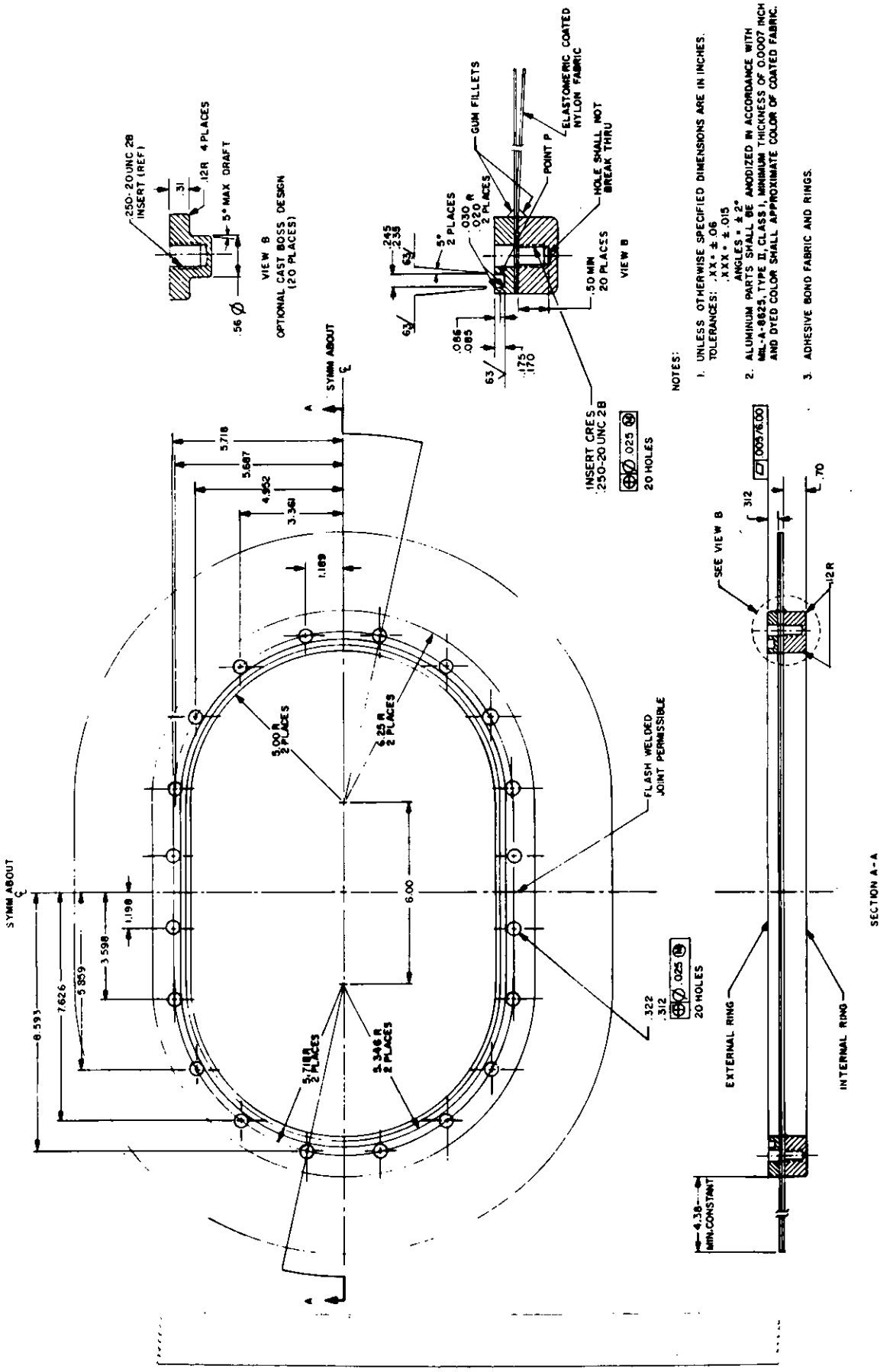
ALL DIMENSIONS ARE IN INCHES

STEEL BRACE, RIGHT SIDE

FIGURE 6. Support braces.

X-1006B

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X-3323D

FIGURE 7. Access door fitting-compression type

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-D-52033D(ME)	Distributor, Water, Collapsible-Tank: 900-Gallon Capacity, with Spraybar Assembly

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*: _____
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

DD FORM 82 MAR 1426 PREVIOUS EDITION IS OBSOLETE.

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